VIMANSHA JAYARATHNA

Software Engineer | AI Engineer

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PROFILE

AI/ML-focused Software Engineer with hands-on experience building LLM-powered features with Gemini APIs, including a virtual fit-on room and styling recommendations, and deploying predictive models into live FastAPI backends. Skilled in Python, TensorFlow, Scikit-learn, RAG architecture, and LangChain, with full-stack ability in Next.js and PostgreSQL. Strong statistical and optimization background from a BSc (Hons) in Statistics & Operations Research.

SKILLS

Languages & Query Tools — SQL, Python, JavaScript, TypeScript, R, Java

Frontend — React, Next.js, Vite, Vue.js, Tailwind CSS, Bootstrap, ShadCN UI, Zustand, Zod, Lovable

Backend — NestJS, Flask, FastAPI, Node.js, REST APIs, JWT, GraphQL

Database — MySQL, PostgreSQL, MongoDB, Snowflake, SQL Server

Generative AI & LLMs — Prompt Engineering, LLM Integration (Gemini API), RAG Architecture, LangChain, Structured Outputs (JSON), Context Engineering

Machine Learning — TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Predictive Modeling, Regression Analysis, Time Series Analysis, Multivariate Analysis, Optimization (PuLP)

Data & Analytics Platforms — Power BI, Tableau, Snowflake

AI-Assisted Development — Claude Code, Codex, Gemini, ChatGPT, Lovable, AI-Assisted Debugging & Code Review

Cloud & DevOps — AWS, Azure, Docker, Linux

Tools — Jira, Git/GitHub, Postman, Swagger, VS Code

Soft Skills — Communication, Teamwork & Collaboration, Analytical Thinking, Customer Centricity, Fast Learning, Adaptability

WORK EXPERIENCE

Insharp Technologies (PVT) Ltd

Associate Software Engineer Apr 2026 – Present

- Contributing full-stack to an ERP systems covering order, customer/supplier, expense, and employee management modules.
- Debugging and resolving issues across frontend and backend to ensure system stability and reliability.
- Working within a cross-functional team using Git/GitHub for version control and Jira for agile project tracking.

Tools & Technologies: NestJS, React, Vite, Tailwind CSS, MongoDB, Git, GitHub, Jira

Intern AI/ML Engineer Oct 2025 – Apr 2026

- Built AI features for a women's clothing platform, including a virtual fit-on roomand LLM-based styling recommendations using Gemini APIs.
- Developed full frontend (Next.js, Tailwind CSS) with backend integration (Python, FastAPI) for client and admin portals.
- Applied predictive modeling (linear regression) and deployed ML components using PostgreSQL, Swagger, and Postman.

Tools & echnologies: Python, Linear Regression, FastAPI, Next.js, Tailwind CSS, PostgreSQL, Swagger, Postman

EDUCATION

BSc(Hons) in Statistics & Operations Research | Second Upper 2021 – 2025

Faculty of Science - University of Peradeniya

Relevant Coursework:

- Probability Theory, Sampling Techniques, Quality Control, Regression Analysis, Multivariate Methods, Time Series Analysis, Stochastic Processes, Statistical Data Mining

- Financial Mathematics, Operations Research, Network Optimization, Manufacturing & Logistics, Linear Algebra, Optimization Modeling
- OOP, Data Structures, Web Programming, Database Management Systems, Design and Analysis of Algorithms, Artificial Intelligence, Neural Networks and Deep Learning

GCE Advanced Level (Physical Science) 2019
Ruwanwella Central College
- Results: BBB
- Z-Score: 1.5006

RESEARCH & PROJECTS

An Optimization Technique for Assembly Line Balancing in Apparel Sector
A Case Study at Brandix Apparel Solutions (Pvt) Ltd - Published Work
- Introduced a method for efficient line layout and developed a mathematical model for MO allocation, achieving a more efficient solution than the current manual method.
- Built a user-friendly web interface for automated MO allocation.
- Analyzed absenteeism trends using time series methods and proposed workforce planning improvements.

Tools & Technologies: Python, PuLP, Streamlit, Pandas, Time Series Analysis

Laptop Price Predictor
- Developed a machine learning-powered web application to predict laptop prices based on user inputs (RAM, brand, OS, CPU), enhancing user decision-making for tech purchases.

Tools & Technologies: Python, Flask, Pandas, NumPy, Scikit-learn

AI-Powered Tweet Classification
- Developed a deep learning-based web interface to classify tweets as "personal health mentions" or "non-personal health mentions" using sequence models.
- Implemented LSTM and Bi-LSTM neural networks, achieving over 80% classification accuracy.

Tools & Technologies: Python, TensorFlow, Keras, Pandas, Scikit-learn, Matplotlib, Gradio, Google Colab

CERTIFICATIONS

Data Warehousing Workshop - Snowflake | Fundamentals of Azure AI services | Version Control | Fundamental AI Concepts | Introduction to Machine Learning | Perform analytics in Power BI

VOLUNTEERING

STEMUP Educational Foundation Aug 2023 – Present
Community Leader

Statistical Circle - University of Peradeniya 2023 – 2024
Society Member

Operations Research Society - University of Peradeniya 2021 – 2022
Committee Member

REWARDS & INVOLVEMENTS

The Rising Star Award 2025
STEMUP Educational Foundation

1st place in Social Services Category 2023
ACES HACKATHON

REFERENCES

Mr. Harshana Dharmawardhana, *Associate Tech Lead,* GTN Tech
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Dr. Niluka Rodrigo, *Senior Lecturer,* Department of Mathematics - University of Peradeniya
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